

ANDREW WAN

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EDUCATION

University of California, Irvine

Masters of Engineering with concentration in Electrical Engineering and Computer Science

Irvine, CA

June 2027

University of California, Los Angeles

Bachelor of Science, Computer Engineering - Minor: Film TV

Los Angeles, CA

March 2026

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Databases, Neural Networks

Languages: Python, , JavaScript, TypeScript, SQL, HTML/CSS, C++, TensorFlow, PyTorch, NumPy, Pandas, Bash

Frameworks/Tools: React, Node.js, MongoDB, Supabase, PostgreSQL, Git, Linux/macOS, Power BI, Excel, REST APIs

Hardware/Embedded: Raspberry Pi, STM32, IMU sensors

Interests: Computers, Volleyball, Video Editing, FPS Games, Guitar, Rock Climbing, Photography, Snowboarding

EXPERIENCE

UCLA IT

BruinCast AV Operations

Los Angeles

November 2024 - Present

- Conduct code reviews for Python **automation scripts** supporting metadata tagging, file management, and upload verification, reducing manual processing time by ~3 hours
- Use BruinPane and classroom-specific networks to **remotely monitor PTZ** recording cameras and webcam feeds, helping **troubleshoot** video and audio issues across 260+ classrooms and courses

Partners Worldwide | World Vision

Senior Consultant, Data Science | Providence Consulting

Los Angeles

February 2024 - July 2025

- Crafted LLM prompt workflows using **context injection**, task and output framing, and **error reduction** to extract trends, key topics, and recurring themes from text-based interviews improving speed and consistency of qualitative analysis
- Wrote **Python** scripts to data clean, normalize, and categorize **thousands** of unstructured data to enable reliable analysis
- Collaborated with **6** contractors and consoled with clients weekly, presenting timely deliverables

Code Ninjas

Instructor

Los Alamitos

November 2024 - June 2025

- Teach students programming fundamentals including, problem solving, logic, and basic software development
- Adapted project support** and lesson plans across different skill levels while **maintaining an engaging** environment

Shortstack

Data Science Intern

New York

June 2024 - August 2024

- Developed **Python scripts** to collect and structure web data for downstream machine learning and document workflows
- Automated retrieval of legal and policy documents from **~1000** websites using **Python and search APIs** building a repeatable data collection workflow that improved consistency and reduced manual collection time by 60%

PikNik

Data Center Intern

San Diego

June 2020 - November 2021

- Assisted with assembly and maintenance of multi-NVIDIA GPU server systems supporting **Web3** infrastructure
- Troubleshoot hardware, **network connectivity**, and system issues in data-center environments to support reliable server deployment and operation of high-performance compute systems

PROJECTS

AirWave | *Capstone - Raspberry Pi, IMU, Pan-Tilt Camera, Spotify, Voice, 3D Printing, MQTT, JavaScript, HTML*

- Designed **MQTT**-based communication between **Raspberry Pi+IMU+Pan-tilt** camera tracking to synchronize sensor and gesture input, voice input, camera tracking and music playback control in real time
- Browser **dashboard** with HTML, CSS, JavaScript, and MQTT over WebSockets to monitor features

Fridayz | *React, Node.js, Express.js, MongoDB, JavaScript, REST APIs, JWT Authentication*

- Co-developed a weekly blogging social web app, collaborating in an agile team from ideation and feature planning
- Implemented core social features including **User authentication**, friend management, and commenting on weekly posts
- Developed **backend APIs** for users and posts and connected them to a React frontend for profile and community features

Building a NAS | *RAID 5, 12TB HDDs, Networking*

- Built a home NAS using TruNAS from spare components and configured 3 HDDs in **RAID 5** for ~10TB usable storage
- Configured **TCP/IP** networking, **SMB** file-sharing, and **SSH** remote administration to enable reliable access and management across the local network
- Set up network shares, access controls, firewall settings, and remote administration **protocols** to centralize backups

OpenClaw | *macOS, LLM APIs, GitHub, Scripting*

- Configured OpenClaw/ClawdBot on a Mac mini to enable an **LLM-assisted automation agent** with root level permissions for executing local workflows and system tasks such as daily texts and stock updates reducing manual effort by 50%
- Implemented a modular command/tool setup to trigger repeatable automations using **APIs** and **LLM tokens**

Simon Game | *IoT, STM32, IMU/Gyro, MachineLearning*

- Built an embedded game inspired by Simon using an **IMU (acceleration + rotation)** to detect and validate user motion
- Programmed an **STM32** microcontroller in STM32CubeIDE and implemented a USB 2.0 wired interface to connect
- Engineered motion extraction from gyroscope/accelerometer signals and applied to a **ML classifier** to infer gesture made